Adding USGS channel measurements to RiverShapeML

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RiverShapeML (Modaresi Rad et al., 2024) predicts bankfull width and depth for all 2.65 million NHDPlus reaches in CONUS from a model trained on HYDRoSWOT, the USGS collection of acoustic Doppler (ADCP) measurements through 2014. USGS now publishes every channel measurement it makes, by every method (wading, bridge, cableway, boat), through its Water Data OGC API: 4.77 million records. This note asks whether those measurements improve the model, with everything else held fixed.

How it started. The USGS gage on Shoal Creek at Silverway Drive in Austin (08156675), at the end of a reach I walked, has 68 width measurements. HYDRoSWOT has 2, both from 2009, so the site falls below the 5-measurement minimum and is not used. The other 66, mostly wading measurements, are simply outside HYDRoSWOT. Across CONUS, that pattern turns out to be common, and most common in small streams.

What I did. I rebuilt the bankfull targets from all USGS measurements, using RiverShapeML's sanity rules and Mike Johnson's AHGestimation for the hydraulic-geometry fits, evaluated at the 2-year flood (the median annual peak). I replaced the $R^2 \geq 0.6$ site filter with a filter on the precision of the target itself, because many well-measured sites have nearly constant width and fail an R^2 test for that reason alone. The features, feature engineering and inference are those of the public NOAA-OWP/predict-riverML pipeline, whose national outputs I could reproduce exactly from the published models; the model is the same XGBoost setup with modest regularization and early stopping.

More sites where they were scarcest. Bankfull-width training sites rise from 4,375 to 6,928 (+58%). Sites draining under 10 km² rise from 66 to 111, and 10–100 km² from 596 to 1,157; 58% of NHDPlus reaches are under 10 km². Where non-ADCP and ADCP measurements were taken the same way at the same site and flow, widths agree within 0.1–2.4%.

Lower error in every size class, most in small streams. I trained the same model twice: on the existing RiverShapeML data (HYDRoSWOT measurements at existing sites) and on the expanded set (all measurements, existing and new sites), and scored both on the same held-out sites with cross-validation grouped by reach.

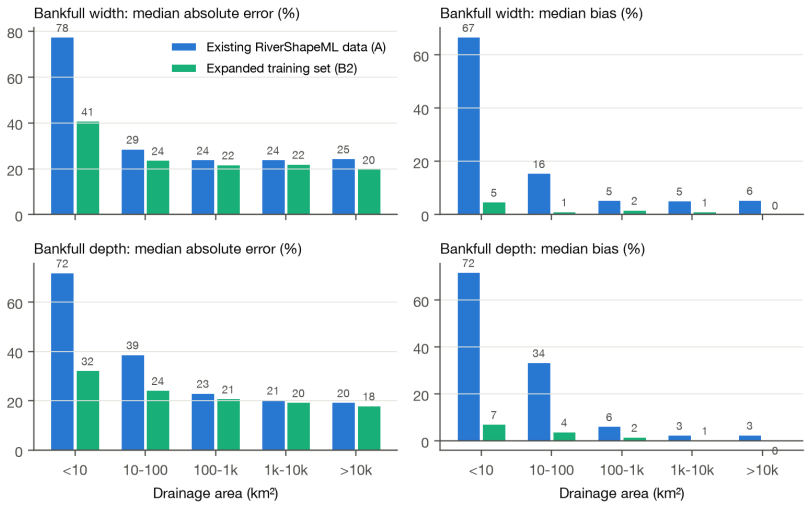


Figure 1. Held-out median absolute error and median bias by drainage area, existing RiverShapeML data (A) vs the expanded training set (B2).

	Width	Depth
Median absolute error, all sites	25.4% → 22.4%	23.8% → 20.9%
Median absolute error, < 10 km²	78% → 41%	72% → 32%
Median bias, < 10 km²	+67% → +5%	+72% → +7%
Median absolute error, sites new to training	32% → 26%	38% → 25%
Existing sites, on the original HYDRoSWOT targets	20.4% → 19.6%	18.2% → 18.2%

The model trained on the existing data overpredicts small streams because it has few to learn from; the new sites remove most of that bias. The gain comes from the new sites; at existing sites, judged on the original HYDRoSWOT targets, the two models match.

A re-predicted national product. I ran the retrained model through the predict-riverML pipeline for every reach, with split-conformal 90% prediction intervals by drainage-area class (coverage 0.88–0.90 on held-out sites). Between 100 and 10,000 km² it agrees with the current model within about 10%. In headwaters it is narrower and closer to the Bieger et al. (2015) regional curves, so the national median width falls from 18.3 m to 10.9 m. Implausibly wide reaches (> 1 km) drop from 0.47% to 0.002%.

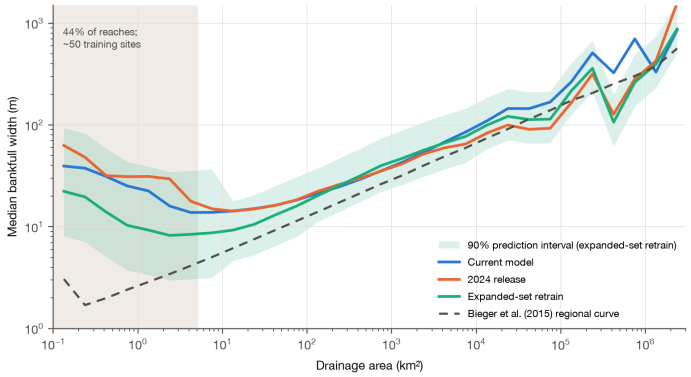


Figure 2. Median bankfull width by drainage area, all CONUS reaches. The shaded region holds 44% of reaches but only ~50 training sites; the gap to the regional curves narrows but does not close.

Questions I would value your view on.

1. Was the HYDRoSWOT-only choice deliberate, for example to keep one measurement method? The agreement above suggests other methods can be added with little risk, but there may be reasons I have not seen.
2. Headwaters still have few sites. Monotone constraints on drainage area, or blending with regional curves below a threshold, could keep predictions on a physical trend where data are thin.
3. About 7,000 more sites have enough measurements but short flow records; NWM-based flood frequencies, which the codebase already supports, could bring them in.
4. The same measurements are much denser at in-channel flows, which maps directly onto the in-channel model.

I would be glad to prototype any of these within predict-riverML.

Credit and materials. This builds directly on Arash Modaresi Rad's RiverShapeML and the NOAA Office of Water Prediction's open predict-riverML pipeline, whose published models and inference code made a controlled comparison possible, and on AHGestimation by Mike Johnson, with contributions from Arash Modaresi Rad. A longer write-up, the code (one script rebuilds everything from public sources), an interactive notebook, and the re-predicted product with intervals for every reach are available on request.

References. Modaresi Rad, A., et al. (2024), *JGR: Machine Learning and Computation*, doi:10.1029/2024JH000173. Bieger, K., et al. (2015), *JAWRA* 51(3), 842–858, doi:10.1111/jawr.12282. Johnson, J. M., et al., AHGestimation (R package).